

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode

• **Backend & Systems:** FastAPI, REST APIs, ETL Pipelines, Kafka, PySpark, gRPC, Protocol Buffers, Grafana, InfluxDB, Kubernetes

• **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Owned end-to-end design and deployment of a cross-platform data integration system ingesting high-frequency sensor data from 100+ users, enabling real-time operational data collection and downstream insight generation for engineering and research teams

• Designed and optimized **PostgreSQL schema, indexing strategies, and query execution plans** for high-volume datasets, improving query performance by 38% under production workloads

• Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by **43%**

• Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed highly **scalable NLP pipelines in Python** to process high-volume datasets, reducing processing latency by **41%**

• Built distributed real-time data ingestion pipelines using Kafka and Neo4j Streams to capture, route, and process high-throughput operational data streams from external systems, enabling downstream pattern detection and engineering insights

• Designed and shipped production-grade REST APIs using FastAPI to integrate data systems across distributed environments, implementing validation, error handling, monitoring, and automated CI/CD deployment workflows enabling reliable cross-system integration at scale

• Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Built and deployed production data integration platform using FastAPI and PostgreSQL, connecting disparate data sources and custom ingestion pipelines to enable real-time querying and operational insight generation across distributed system architectures

• Designed fault-tolerant data pipelines connecting external systems and customer infrastructure, enabling low-latency, reliable data availability for downstream engineering workflows and analytics

• Improved system performance by **64%** through optimized indexing strategies and pipeline efficiency improvements in production environments

• Developed backend services enabling real-time, high-performance querying of unstructured data across distributed system architectures

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving visibility into operational performance across transportation services and decision-making insights

• Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure patterns, trends, and performance issues in distributed systems environments

• Built custom Python scripts and SQL tooling to integrate, validate, and analyze operational data across customer systems, mapping dependencies and surfacing data quality issues that accelerated root-cause analysis by 26%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Deployed and maintained 20+ custom reporting and analytics solutions across enterprise client environments in Sales, Finance, and Operations, integrating data from customer systems using SQL and DAX to enable operational visibility and business decision-making

• Automated recurring **Excel**-based reporting workflows using **VBA**, implementing data validation, transformation, and scheduling logic, reducing report preparation time by **42%** and improving data consistency across weekly reporting cycles

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of 6 engineers and designers to ship product updates for DeepCover, using telemetry data, user feedback, and performance analysis to prioritize engineering decisions and guide development

• Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog grooming, sprint planning, and issue resolution to improve cross-team visibility and ensure on-time milestone delivery

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

• Built an **end-to-end machine learning system** using Python, Pandas, and LightGBM to forecast customer spending and assess credit risk from large-scale **time-series** transaction data, including data preprocessing, feature engineering, model training, and evaluation

• Designed and implemented **scalable ETL and feature engineering pipelines**, capturing behavioral trends